

**Flame Watch Smart Mobile Rover for Real-Time Fire
Detection, Alarm Activation, Mobile Alerts, Bluetooth
Navigation, and Camera-Based Environmental Monitoring**

A PROJECT REPORT

Submitted by

JAIYANTH S K (113222071035)

MITTHUN G (113221071059)

in partial fulfillment for the award of the degree

of

BACHELOR OF TECHNOLOGY

IN

INFORMATION TECHNOLOGY



VELAMMAL ENGINEERING COLLEGE, CHENNAI - 66.

(An Autonomous Institution, Affiliated to Anna University, Chennai)

APRIL 2025

VELAMMAL ENGINEERING COLLEGE

CHENNAI-66



BONAFIDE CERTIFICATE

Certified that this Project report “**Flame Watch Smart Mobile Rover for Real-Time Fire Detection, Alarm Activation, Mobile Alerts, Bluetooth Navigation, and Camera-Based Environmental Monitoring**” is the bonafide work of **JAIYANTH S K** and **MITTHUN G** who carried out at the project work under my supervision.

PROFESSOR & HEAD

Dr. JEEVAA KATIRAVAN

Head of the Department

Information Technology

Velammal Engineering College

Ambattur Red Hills Rd,

Surapet, Chennai- 600066.

SUPERVISOR

Mrs. A.R.SATHYABAMA

Assistant Professor

Information Technology

Velammal Engineering College

Ambattur Red Hills Rd,

Surapet, Chennai- 600066.

CERTIFICATE OF EVALUATION

COLLEGE NAME : VELAMMAL ENGINEERING COLLEGE

BRANCH : INFORMATION TECHNOLOGY

SEMSETER : VIII

Sl. No	Name of the students who has done the project	Title of the Project	Name of supervisor with designation
1	Jaiyanth S K	Flame Watch Smart Mobile Rover For Real Time Fire Detection, Alarm	Mrs. A.R.Sathyabama Assistant Professor
2	Mitthun G	Activation, Mobile Alerts, Bluetooth Navigation, and Camera Based Environmental Monitoring	

This report of project work submitted by the above students in partial fulfillment for the award of Bachelor of Technology Degree in Anna University was evaluated and confirmed to be reports of the work done by the above students and then assessed.

Submitted for Viva voce Evaluation held on.....

Internal Examiner

External Examiner

ABSTRACT

Fires at homes, factories, or woods can hurt people, break things, or harm nature. Spotting flames fast - then reacting right away - helps reduce injuries, lost trees, or ruined buildings. But most older alarm setups stay fixed in one place, cover only small zones, also miss live updates on what's happening around them. Because of that, rescue teams might arrive late, making bad moments even worse. To tackle these issues, this project introduces Flame Watch - a compact rover powered by IoT that spots fires early using smart tech. Instead of just one method, it uses flame sensors along with smoke detectors so mistakes happen less often. A tiny computer inside checks incoming sensor info nonstop, helping the system react quickly once danger levels cross a set limit. When it senses heat or smoke, Flame Watch turns on a loud alarm to alert people close by while shooting alerts to linked phones using wireless signals. Instead of walking into danger, folks can steer the rover from afar thanks to its built-in Bluetooth remote drive. That way, they stay out of harm's reach when dealing with risky spots tied to fires. A built-in high-res camera sits on the rover, sending live video straight from its location. Because of this stream, controllers can judge how bad fires are while also watching changes in surroundings - helping them act fast when things go wrong. Since the unit moves around freely, it checks more ground than regular static alarms do, so it fits well in places like storage hubs, factories, or wooded regions. The Flame Watch setup keeps costs low, uses

little power, while growing smoothly when needed - so it works well whether you're protecting a single building or an entire city. Down the line, smarter upgrades could bring AI-powered fire type detection that learns over time, self-guided movement through tricky spaces by sensing barriers, exact positioning via GPS to speed up rescue alerts, also live data uploads to online platforms for spotting risks before they happen. Thanks to such features, this tool might grow beyond basic monitoring into a full emergency network that boosts how we handle flames and get ready for crises. Overall, Flame Watch demonstrates how early fire response and situational awareness can be greatly enhanced by combining mobility, real-time sensing, and wireless communication. This strategy lowers human risk while facilitating quicker and better-informed emergency responses, highlighting a workable route toward safer environments.

Keywords : Finding fire, gadgets that talk to each other, remote-controlled vehicle on wheels, tool that sees flames, device that catches smoke, instant warning setup, wireless joystick control, watching through live video feed, clever protection tech, quick help during crises.

ACKNOWLEDGEMENT

We are greatly and profoundly thankful to our honorable Chairman, **Shri. M.V. Muthuramalingam**, for facilitating this opportunity. We also record our sincere thanks to our Chief Executive Officer, **Mr. M.V.M. Velmurugan**, and our Deputy Chief Executive Officer, **Mr. V. Karthik Muthuramalingam**, for extending a generous hand in providing the best resources of the college. We are thankful to our Principal **Dr. S. Satish Kumar**, for rendering moral support to us during the course of our project. We are also thankful to our Head of the Department **Dr. Jeevaa Katiravan**, for the supports and useful suggestions during this project work. We thank our project guide **Mrs. A.R.Sathyabama** for her constant technical support and encouragement, which enabled us to complete our project successfully. We would like to express our sincere gratitude to our project coordinator **Mrs. A.R. Sathyabama & Mrs. S.P. Revathy** for her moral support throughout this project. We are also greatly indebted to the faculty members of the Information Technology Department for their coordination and support in helping us to finish this project. We would also like to take this opportunity to thank all our family members, classmates, and friends who offered an unflinching moral support for completion of this project. The success and final outcome of this project required a lot of guidance and assistance from many people and we are extremely privileged to have got this all along the completion of our project.

TABLE OF CONTENTS

Sl. NO	TITLE	PAGE NO
1	Abstract	IV
2	Acknowledgement	V
3	Table of Content	VI
4	List of Figures	VII

TABLE OF CONTENTS

CHAPTER NO	TITLE	PAGE NO
01	INTRODUCTION	01
	1.1 Introduction to Fire Detection Systems	02
	1.2 Smart Mobile Rover Concept	02
	1.3 Flame Sensor and Fire Detection Mechanism	03
	1.4 GSM Module for Alert Message	03
	1.5 Bluetooth Module for Navigation	03
	1.6 Camera-Based Environmental Monitoring	04
	1.7 Buzzer Alarm System	04
02	RELATED WORK	05
03	METHODOLOGY	07
	3.1 Proposed Work Methodology	07
	3.1.1 Hardware Components	08
	3.1.2 Software Implementation	12
	3.1.3 System Architecture	13
	3.1.4 Algorithm Steps	20
04	RESULT AND DISCUSSION	22
	4.1 Software and Hardware Requirements	22
	4.1.1 Software Requirements	22
	4.1.2 Hardware Requirements	25
	4.2 Result	29
	4.3 Performance Analysis	35
	4.4 Discussion	36
05	CONCLUSION AND FUTURE WORK	41
06	CODING	45
07	REFERENCES	59

LIST OF FIGURES

Fig. No	Title	Pg. No
3.1	Overall System Architecture of Flame Watch Rover	17
3.2	Block Diagram of Fire Detection and Alert System	18
3.3	Working Flow of GSM-Based Alert Message Transmission	18
3.4	Bluetooth Navigation Control Process	19
3.5	Camera-Based Live Monitoring Setup	19
4.1	Flame Sensor Detection Module	30
4.2	GSM Module for Mobile Alert Notification	31
4.3	Bluetooth Module Interface with Smartphone	31
4.4	Buzzer Alarm Activation Circuit	32
4.5	Mobile Application Alert Message Screenshot	32
4.6	Live Camera Monitoring Display on Smartphone	33
4.7	Rover Movement Control via Bluetooth	33
4.8	Complete Hardware Assembly of Flame Watch Mobile Rover	34
4.9	Fire Detection and Alarm Activation Testing	34
4.10	Performance Analysis and Response Time Comparison Chart	36

CHAPTER-1

1.INTRODUCTION

Fire accidents pose a serious threat to human life, property, forests, industries, and public infrastructure. Delayed detection and response to fire outbreaks can lead to irreversible damage and financial loss. Conventional fire monitoring systems are generally fixed in one location and lack mobility, real-time monitoring, and smart alert capabilities. To overcome these limitations, the proposed system Flame Watch Smart Mobile Rover for Real-Time Fire Detection, Alarm Activation, Mobile Alerts, Bluetooth Navigation, and Camera-Based Environmental Monitoring is developed. The Flame Watch Smart Mobile Rover is an intelligent robotic system designed to detect fire in real time using flame sensors and immediately alert users through GSM-based mobile notifications. The system is equipped with a buzzer alarm for instant local alert activation and a live camera module for environmental monitoring. The rover can be controlled via Bluetooth navigation using a smartphone, allowing users to remotely monitor and move the system toward suspicious areas. This smart rover integrates embedded systems, wireless communication, and IoT concepts to provide a portable, efficient, and cost-effective fire monitoring solution. The proposed system ensures early detection, rapid alert transmission, and live surveillance, thereby reducing response time and minimizing damage caused by fire hazards.

1.1 Introduction to Fire Detection Systems

Fire detection systems are designed to identify fire hazards at an early stage and provide alerts to prevent damage and loss. Conventional systems typically use smoke sensors, heat sensors, or flame sensors to detect abnormal environmental conditions. However, most traditional systems are stationary and do not provide remote access or mobility. The Flame Watch Smart Mobile Rover enhances fire detection by combining real-time sensing with wireless communication and robotic mobility, enabling faster and more efficient emergency response. .

1.2 Smart Mobile Rover Concept

The Smart Mobile Rover is a robotic vehicle equipped with sensors and communication modules. Unlike fixed fire alarm systems, this rover can move toward different locations to inspect fire-prone areas. The rover is controlled wirelessly through a smartphone using a Bluetooth module. This mobility allows users to monitor large areas such as warehouses, industries, laboratories, or forest zones. The integration of detection, alerting, and navigation makes the system intelligent and highly effective.

1.3 Flame Sensor and Fire Detection Mechanism

The flame sensor is the core component responsible for detecting fire. It senses infrared radiation emitted by flames within a specific wavelength range. When the sensor detects fire intensity above a predefined threshold. A signal is sent to the microcontroller. The system confirms fire detection. Emergency actions are

triggered immediately. This mechanism ensures fast and accurate fire identification before it spreads extensively.

1.4 GSM Module for Alert Message

The GSM module is responsible for sending alert notifications to the user's mobile phone. Once fire is detected the microcontroller sends a command to the GSM module. SMS alert message is automatically sent to the registered phone number. The message indicates fire detection and warning information. This feature ensures that the user is informed instantly, even if they are far from the location.

1.5 Bluetooth Module for Navigation

The Bluetooth module enables wireless communication between the rover and a smartphone. Using a mobile application. The user can move the rover forward, backward, left, and right. The rover can be directed toward the fire location. Inspection of hazardous areas can be performed remotely. This navigation capability increases flexibility and safety during emergency situations.

1.6 Camera-Based Environmental Monitoring

When connected to the smartphone:

- Users can monitor the affected area visually.
- Fire intensity and spread can be observed.
- Environmental conditions can be assessed remotely.

This feature enhances situational awareness and supports better decision-making during fire emergencies.

CHAPTER 2

2.1 RELATED WORK

In recent years, several smart fire detection and surveillance systems have been developed to reduce fire-related hazards using embedded systems and wireless communication technologies. Flame sensors, temperature sensors, smoke detectors, and microcontroller-based systems such as Arduino Uno and Raspberry Pi are widely used for early fire detection and automated alert mechanisms. These systems enhance safety by continuously monitoring environmental conditions and triggering alarms when fire is detected [1].

GSM-based fire alert systems have been proposed to provide real-time notifications to users. By integrating GSM modules like SIM800L GSM Module, the system sends SMS alerts directly to registered mobile numbers when flame or smoke is detected. This approach ensures immediate remote notification even when users are away from the monitoring location, thereby minimizing damage and improving emergency response time [2].

Bluetooth-controlled robotic systems have also been introduced for navigation and surveillance in hazardous environments. Modules such as HC-05 Bluetooth Module enable wireless control of mobile rovers through smartphone applications. These systems allow users to remotely navigate robots into fire-prone or restricted areas, ensuring human safety during inspection and monitoring tasks [3].

Camera-based environmental monitoring systems using modules like ESP32-CAM

provide live video streaming capabilities. Such systems allow real-time visual monitoring of the affected area through mobile devices. Live camera feeds assist in assessing fire intensity, identifying the source, and planning immediate control measures efficiently [4].

Buzzer-based alarm systems are integrated into fire detection units to provide immediate local alerts. When flame sensors detect fire, the microcontroller activates a buzzer alarm to warn nearby individuals. This dual-alert mechanism (local buzzer and remote GSM alert) significantly improves emergency response effectiveness [5]. Recent advancements combine flame sensors, GSM modules, Bluetooth navigation, and camera-based monitoring into a single smart mobile rover platform. These integrated systems address the limitations of traditional stationary fire alarms by enabling mobility, remote access, real-time alerts, and live environmental monitoring. The Flame Watch Smart Mobile Rover builds upon these technologies to provide an efficient, reliable, and real-time fire detection and alert system suitable for industrial, residential, and smart building applications [6]. This dual-alert mechanism (local buzzer and remote GSM alert) significantly improves emergency response effectiveness [7]. Recent advancements combine flame sensors, GSM modules, Bluetooth navigation, and camera-based monitoring into a single smart mobile rover platform. These integrated systems address the limitations of traditional stationary fire alarms by enabling mobility, remote access, real-time alerts, and live environmental monitoring.

CHAPTER 3

3.1 PROPOSED WORK METHODOLOGY

The proposed system, Flame Watch Smart Mobile Rover, is designed for real-time fire detection, alert generation, and remote monitoring. The rover continuously monitors the environment using a flame sensor and transmits alerts through GSM while allowing live navigation and camera monitoring through Bluetooth. The methodology integrates hardware components (sensors, microcontroller, communication modules) with embedded software logic to ensure early fire detection, immediate alarm activation, and secure communication to the user's mobile device.

When fire is detected, the system:

- Activates a buzzer alarm for nearby warning.
- Sends an SMS alert to the registered mobile number using GSM.
- Enables live monitoring through a camera module.
- Allows user navigation control via Bluetooth connection.
- This integrated approach ensures rapid response, mobility, and real-time monitoring in hazardous environments.

3.1.1 HARDWARE COMPONENTS

1. Microcontroller (Arduino Uno Controller)

The Arduino Uno serves as the central processing unit and control hub of the Flame Watch Smart Mobile Rover. It is responsible for receiving input signals from the flame sensor and making logical decisions based on the programmed instructions. When fire is detected, the microcontroller activates the buzzer alarm, triggers the GSM module to send an alert message, and coordinates other response mechanisms. It also manages communication between different hardware modules such as the Bluetooth module for navigation control and the motor driver for rover movement. With its ATmega328P microcontroller, digital and analog input/output pins, and easy programmability using the Arduino IDE, it ensures smooth integration and real-time performance of all system components.

2. Flame Sensor Module

The flame sensor module plays a crucial role in fire detection by sensing infrared (IR) radiation emitted by flames. It is highly sensitive to specific wavelengths of light typically produced during combustion. The sensor continuously monitors the surrounding environment, and when it detects fire, it sends signal to the microcontroller. Hence highly sensitive to specific wavelengths of light typically produced during combustion and flame sensor module plays a crucial role in fire detection by sensing infrared (IR) radiation emitted by flames.

3. GSM Module (SIM800)

The SIM800 GSM module enables long-distance communication by sending SMS alerts to a predefined mobile number when fire is detected. It operates using a SIM card and connects to the cellular network to transmit messages. Once triggered by the microcontroller, the GSM module sends an emergency notification containing information about fire detection. This ensures that users are informed even if they are far away from the monitoring location. The module communicates with the Arduino using serial communication (UART protocol), making it reliable and efficient for real-time alert systems. Its integration significantly enhances the safety aspect of the rover by providing instant remote alerts.

4. Bluetooth Module (HC-05)

The HC-05 Bluetooth module provides wireless communication between the rover and a smartphone. It allows the user to remotely control the rover's movement using a mobile application. Through Bluetooth connectivity, navigation commands such as forward, backward, left, and right are transmitted to the microcontroller. This enables the rover to reach specific areas for inspection and monitoring. The module operates on serial communication and pairs easily with Android devices. Its short-range wireless capability makes it suitable for indoor fire surveillance and controlled navigation environments. Through Bluetooth connectivity, navigation commands such as forward, backward, left, and right are transmitted to the microcontroller.

5. Camera Module (ESP32-CAM)

The ESP32-CAM module provides real-time video streaming and environmental monitoring. It captures live footage of the surroundings and allows users to monitor the situation remotely through a mobile device. Equipped with Wi-Fi capability and an integrated camera, the module can stream video over a network, offering visual confirmation of fire incidents. This feature enhances situational awareness and helps users assess the severity of the fire before taking action. The compact design and built-in processing capability make it efficient for IoT-based surveillance applications. By combining live video monitoring with fire detection, the system becomes more reliable and responsive.

6. Buzzer Alarm System

The buzzer alarm system acts as an immediate local warning mechanism. When the flame sensor detects fire, the microcontroller activates the buzzer to produce a loud audible sound. This alert notifies people nearby about the presence of fire, allowing them to take quick preventive action. The buzzer is simple yet highly effective in emergency situations, especially in environments where visual monitoring may not be constant. It ensures that even without checking mobile notifications, nearby individuals can respond promptly to potential hazards. It alert notifies people nearby about the presence of fire, allowing them to take quick preventive action. The buzzer is simple yet highly effective in emergency situations, especially in environments where visual monitoring may not be constant.

7. Motor Driver (L298N)

The L298N motor driver module controls the movement of the DC motors by regulating their direction and speed. It acts as an interface between the microcontroller and the motors, as the Arduino alone cannot supply sufficient current to drive them. Using the H-bridge configuration, the motor driver enables forward, backward, left, and right movements of the rover. It receives control signals from the microcontroller based on Bluetooth commands and adjusts motor operation accordingly. This ensures smooth and precise navigation of the rover during fire monitoring operations.

8. DC Motors and Wheels

The DC motors and wheels provide mobility to the rover, enabling it to move across different surfaces. The motors convert electrical energy into mechanical motion, allowing the rover to navigate toward fire-prone or suspicious areas. Paired with the motor driver, they support directional movement and speed control. The wheels are designed to provide stability and balance, ensuring smooth movement even in slightly uneven environments. This mobility feature makes the system dynamic and capable of reaching areas where fixed fire detection systems cannot operate. It with the motor driver, they support directional movement and speed control. The wheels are designed to provide stability and balance, ensuring smooth movement even in slightly uneven environments. to provide stability and balance, ensuring smooth movement even in slightly uneven environments.

9. Power Supply Unit

The power supply unit consists of a rechargeable battery pack that provides the necessary voltage and current to operate all hardware components. It ensures uninterrupted functioning of the microcontroller, sensors, communication modules, camera, and motors. Proper voltage regulation is maintained to protect sensitive components from fluctuations. A reliable power source is essential for emergency systems, as failure during critical moments can lead to severe consequences. The rechargeable design also makes the rover portable and suitable for long-duration monitoring tasks.

3.1.2 SOFTWARE IMPLEMENTATION

The software implementation includes embedded programming and mobile-side communication.

- Embedded Programming (Arduino IDE)
- Sensor data acquisition.
- Fire detection logic implementation.
- GSM message transmission control.
- Bluetooth command decoding.
- Buzzer activation control.
- Mobile Application and Bluetooth Terminal App
- Sends navigation commands to rover.
- Receives alert notifications.

- Camera Streaming Configuration
- Live video transmission using ESP32-CAM web server or similar module.
- IP-based access for monitoring.

3.1.3 SYSTEM ARCHITECTURE

The system architecture of the Flame Watch Smart Mobile Rover for Real-Time Fire Detection is designed using a layered approach to ensure efficiency, reliability, and real-time responsiveness. Each layer performs a specific function, working together to detect fire, process information, communicate alerts, and enable user monitoring and control. The architecture is divided into four major layers: Sensing Layer, Processing Layer, Communication Layer, and Alert & Monitoring Layer.

The system architecture consists of four main layers:

1. Sensing Layer

The Sensing Layer is the first and most critical part of the system, responsible for continuously monitoring the environment for the presence of fire. It primarily consists of a flame sensor module that detects infrared radiation emitted by flames. The sensor is highly sensitive to specific wavelengths of light associated with fire and can detect flames within a defined range and angle. It provides either analog or digital signals depending on the intensity of the detected flame. This layer operates continuously while the rover patrols the area, ensuring real-time environmental monitoring. By constantly scanning for abnormal heat or flame signatures, the sensing layer acts as the system's eyes, enabling early fire detection.

2. Processing Layer

The Processing Layer acts as the brain of the system and is typically implemented using a microcontroller such as Arduino Uno. This layer receives input signals from the flame sensor and processes the data using programmed decision-making logic. The microcontroller analyzes whether the detected signal exceeds a predefined threshold value to confirm the presence of fire. It also filters out noise and false triggers caused by ambient light or temporary disturbances. Once the detection is validated, the microcontroller executes a series of predefined actions, such as activating the buzzer and triggering communication modules. This layer ensures accurate decision-making, coordinated control of all hardware components, and smooth overall system operation.

3. Communication Layer

The Communication Layer enables remote information transfer and user interaction with the rover. It integrates multiple modules to ensure effective communication. The GSM module is responsible for sending SMS alert messages to the user's registered mobile number immediately after fire detection. This ensures that the user is informed even if they are far from the monitored location. The Bluetooth module allows the user to manually navigate and control the rover using a smartphone application, enabling directional movement and real-time adjustments. Additionally, the camera module captures and transmits live video footage of the environment, allowing the user to visually verify the fire situation.

4. Alert & Monitoring Layer

The Alert & Monitoring Layer focuses on providing both local and remote alerts while enabling continuous surveillance. When the system confirms fire detection, a buzzer alarm is immediately activated to provide an audible warning in the nearby area. This helps alert individuals present at the location to take immediate action. Simultaneously, the user receives an SMS notification through the GSM network, ensuring remote awareness. The live video feed from the camera can be monitored on a smartphone, allowing the user to assess the severity of the fire and make informed decisions. This layer enhances safety by combining instant local alerts with remote monitoring features, ensuring rapid response and improved fire management.

System Flow:

The overall system operates in a structured and sequential manner. Initially, the rover patrols the designated area either autonomously or under user control. During movement, the flame sensor continuously scans the surroundings for fire signals. If a flame is detected, the sensor sends data to the microcontroller for verification. After validating the detection, the microcontroller activates the buzzer alarm to warn nearby individuals. At the same time, the GSM module sends an SMS alert to the user's mobile phone. The user can then connect to the rover via Bluetooth to navigate it closer to the fire source if required. Meanwhile, the onboard camera provides a live video feed to the smartphone, allowing real-time monitoring and better decision-making. This integrated workflow ensures quick detection, immediate alert

generation, and effective remote supervision.

1. Rover patrols the area.
2. Flame sensor detects fire.
3. Microcontroller validates detection.
4. Buzzer alarm activates.
5. GSM module sends alert SMS.
6. User connects via Bluetooth to navigate rover.
7. Camera provides live monitoring feed.

The flame sensor continuously scans the surroundings for fire signals. If a flame is detected, the sensor sends data to the microcontroller for verification. After validating the detection, the microcontroller activates the buzzer alarm to warn nearby individuals. At the same time, the GSM module sends an SMS alert to the user's mobile phone. The user can then connect to the rover via Bluetooth to navigate it closer to the fire source if required. Meanwhile, the onboard camera provides a live video feed to the smartphone, allowing real-time monitoring. Rover patrols the area. Flame sensor detects fire and Microcontroller validates detection then Buzzer alarm activates then GSM module sends alert SMS and User connects via Bluetooth to navigate rover and Camera provides live monitoring feed. This helps alert individuals present at the location to take immediate action. Simultaneously, the user receives an SMS notification through the GSM network, ensuring remote awareness. decisions.

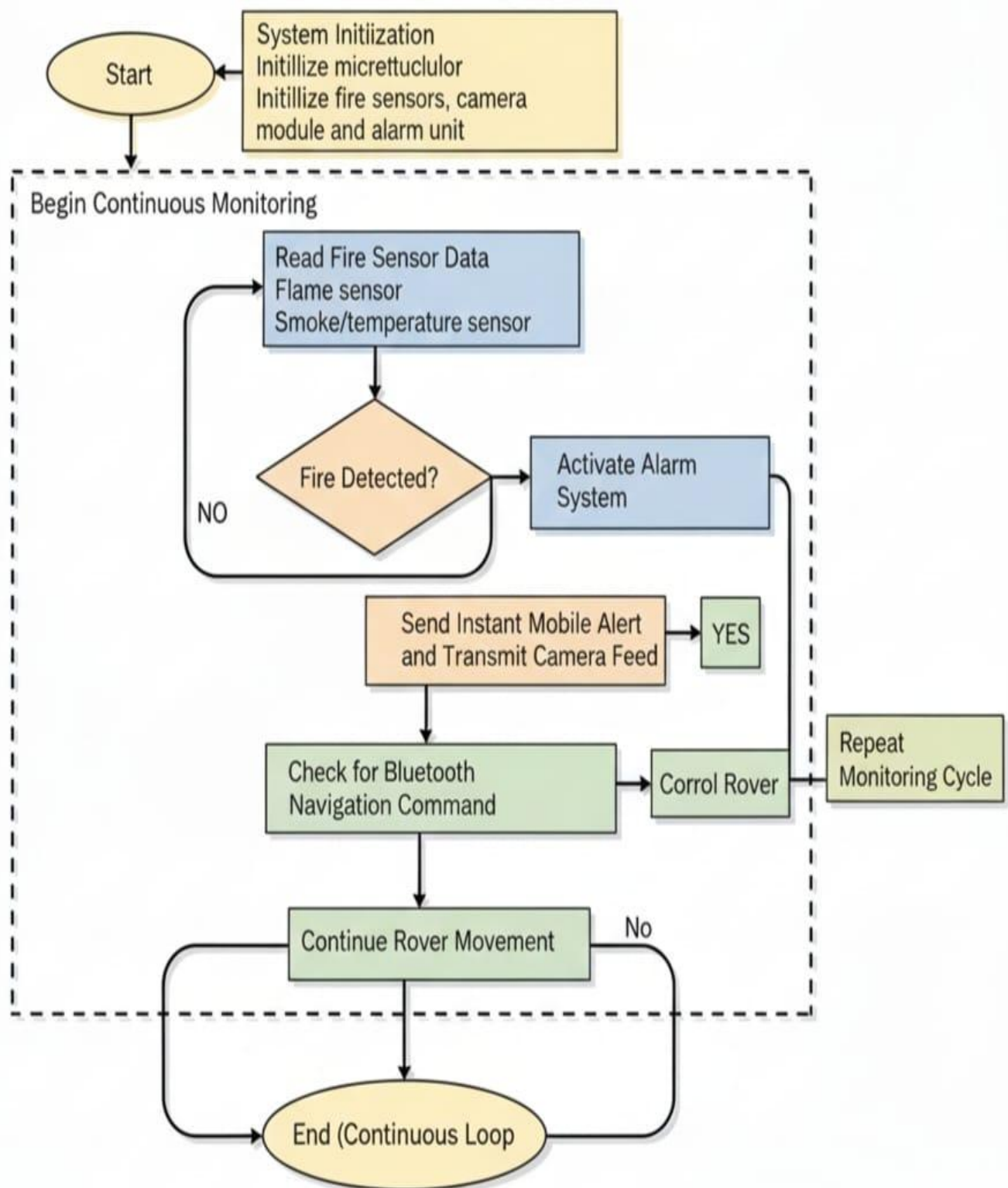


Fig 3.1 Overall System Architecture of Flame Watch Rover

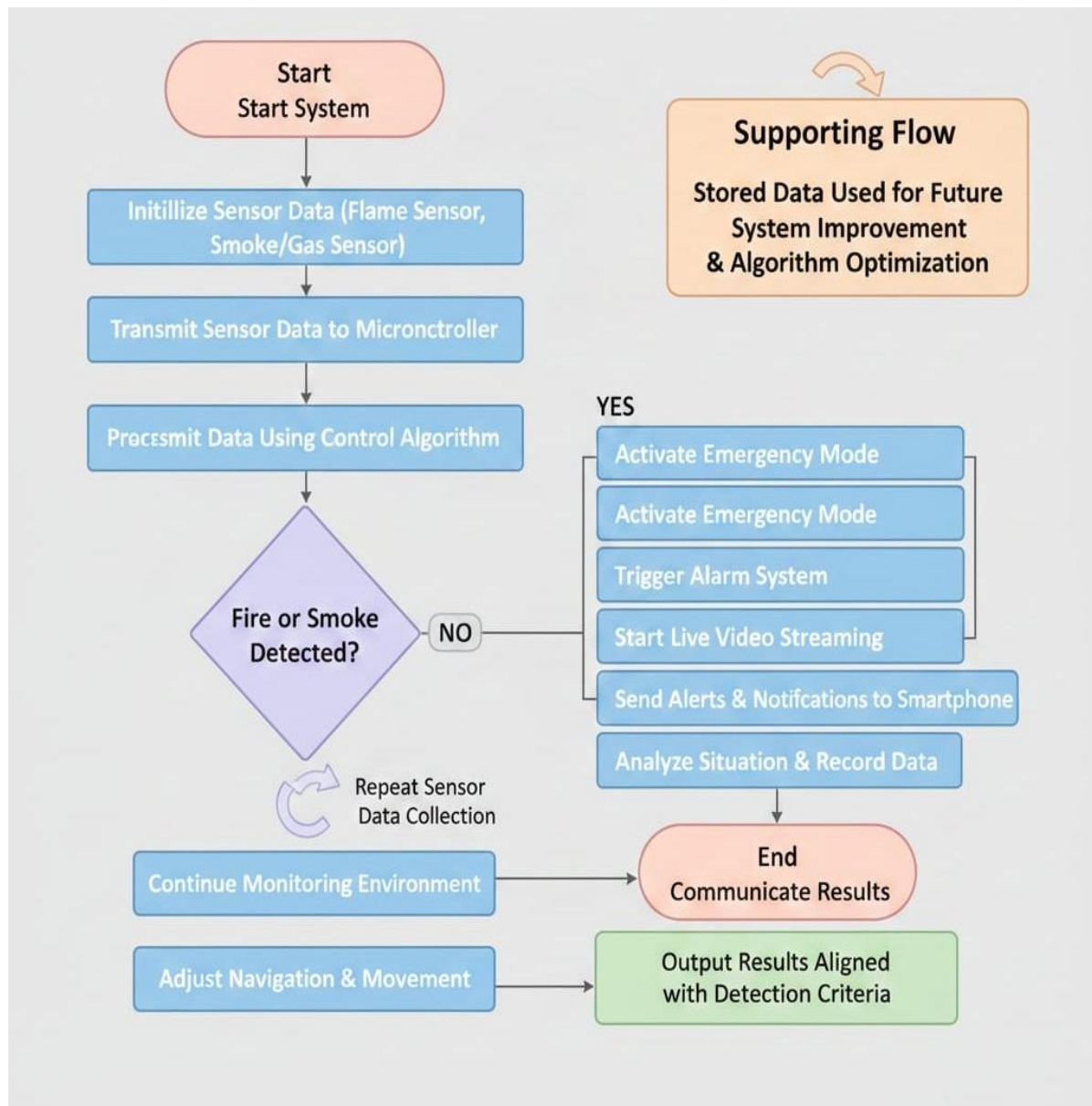


Fig 3.2 Block Diagram of Fire Detection and Alert System

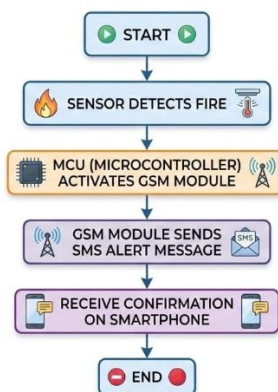


Fig 3.3 Working Flow of GSM-Based Alert Message Transmission

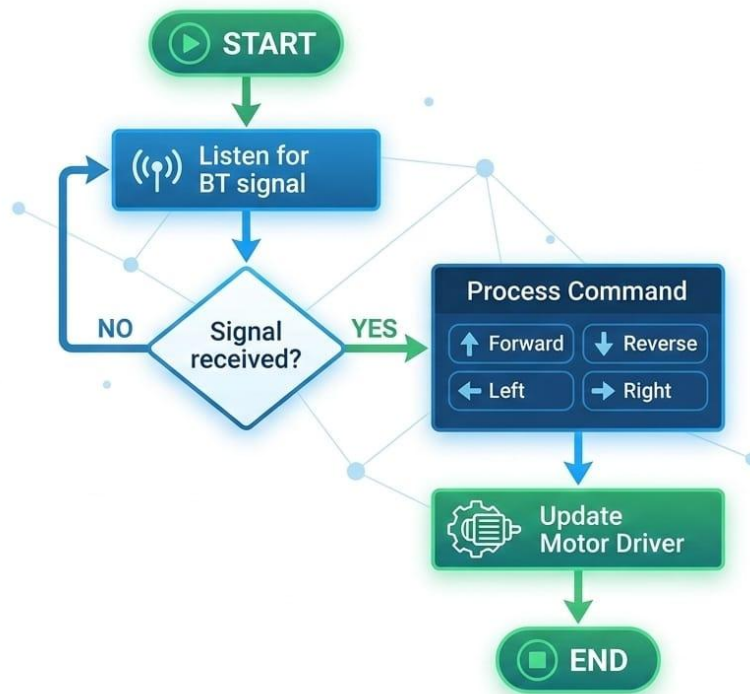


Fig 3.4 Bluetooth Navigation Control Process

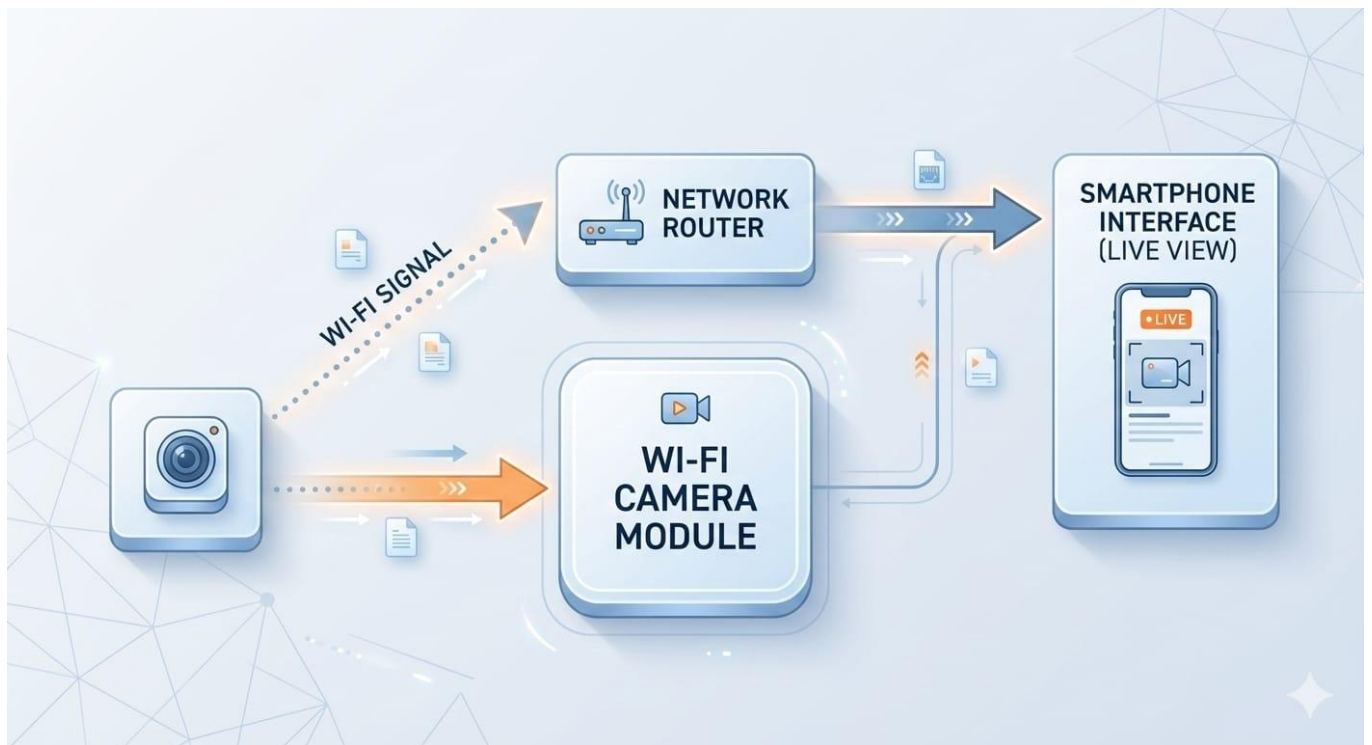


Fig 3.5 Camera-Based Live Monitoring Setup

3.1.4 ALGORITHM STEPS

Step 1: System Initialization

Power ON system.

Initialize microcontroller, GSM, Bluetooth, camera, and flame sensor.

Step 2: Sensor Monitoring

Continuously read flame sensor output.

Step 3: Fire Detection Condition

If flame sensor value > threshold:

Fire Detected = TRUE

Else: Continue monitoring.

Step 4: Alert Activation

If Fire Detected = TRUE:

1. Activate buzzer.
2. Send SMS using GSM module: "Fire detected "
3. Enable camera streaming for live monitoring.

Step 5: Remote Navigation

User connects mobile via Bluetooth.

Send movement commands:

F → Forward

B → Backward

L → Left

R → Right

S → Stop

Microcontroller controls motors accordingly.

Step 6: Continuous Monitoring

Keep checking sensor values.

If fire persists, maintain alert state.

If no fire detected, return to patrol mode.

This helps alert individuals present at the location to take immediate action. Simultaneously, the user receives an SMS notification through the GSM network, ensuring remote awareness. The live video feed from the camera can be monitored on a smartphone, allowing the user to assess the severity of the fire and make informed decisions. This helps alert individuals present at the location to take immediate action.

Simultaneously, the user receives an SMS notification through the GSM network, ensuring remote awareness. The live video feed from the camera can be monitored on a smartphone, allowing the user to assess the severity of the fire and make informed decisions. This helps alert individuals present at the location to take immediate action. Simultaneously, the user receives an SMS notification through the GSM network, ensuring remote awareness. The live video feed from the camera can be monitored on a smartphone.

CHAPTER 4

4.RESULT AND DISCUSSION

4.1 Software and Hardware Requirements

1. 4.1.1 Software Requirements

a. Embedded C

Requirement: Code development and uploading to the microcontroller

The Arduino IDE serves as the primary development environment for writing, compiling, and uploading Embedded C code into the Arduino Uno.

It provides a user-friendly interface that simplifies embedded programming even for complex hardware integration tasks. The IDE includes built-in libraries that support communication protocols such as UART, SPI, and I2C, which are used to interface with modules like GSM, Bluetooth, and camera components. It also offers a serial monitor feature that helps in debugging and testing by displaying real-time sensor readings and system messages. Developers can monitor flame sensor values, check GSM responses, and verify Bluetooth communication during system development. The Arduino IDE streamlines firmware updates and modifications, ensuring that improvements or bug fixes can be quickly implemented and tested. This helps alert individuals present at the location to take immediate action. Simultaneously, the user receives an SMS notification through the GSM network, ensuring remote awareness.

b. Arduino IDE

Requirement: Wireless monitoring and rover control

A mobile application is used as the user interface for wireless monitoring and control of the rover. The app connects to the Bluetooth module installed in the rover and enables real-time navigation commands such as forward, backward, left, and right movement. It can be a generic Bluetooth terminal app or a custom-developed Android application designed specifically for the system. Through this interface, users can remotely operate the rover in hazardous areas without direct exposure to danger. Additionally, the mobile platform allows monitoring of fire alerts and, when integrated with camera streaming, enables live environmental observation. This enhances user safety and situational awareness by allowing immediate action based on real-time visuals and alerts received on the smartphone.

c. Mobile Application (Bluetooth Terminal and Custom App)

Requirement: Sending SMS alerts

GSM network support is essential for enabling long-distance communication and emergency alert transmission. When the flame sensor detects fire, the microcontroller sends AT commands to the GSM module, which then transmits an SMS alert to a predefined mobile number. This ensures that the user receives immediate notification even if they are not within Bluetooth range. GSM communication increases the reliability of the system by

providing remote alerts regardless of physical proximity to the rover. The effectiveness of this feature depends on the availability of cellular network coverage and an active SIM card with messaging services enabled. This capability makes the system suitable for deployment in warehouses, industries, laboratories, and remote environments where continuous monitoring is required.

d. GSM Network Support

Requirement: Sending SMS alerts

GSM network support is essential for enabling long-distance communication and emergency alert transmission. When the flame sensor detects fire, the microcontroller sends AT commands to the GSM module, which then transmits an SMS alert to a predefined mobile number. This ensures that the user receives immediate notification even if they are not within Bluetooth range. GSM communication increases the reliability of the system by providing remote alerts regardless of physical proximity to the rover. The effectiveness of this feature depends on the availability of cellular network coverage and an active SIM card with messaging services enabled. This capability makes the system suitable for deployment in warehouses, industries, laboratories, and remote environments where continuous monitoring is required .

4.1.2 Hardware Requirements

a) Microcontroller (Arduino Uno)

The Arduino Uno acts as the central processing unit of the entire system. It is responsible for reading sensor inputs, processing decision-making logic, and controlling all output devices and communication modules. The board is based on the ATmega328P microcontroller and provides multiple digital and analog input/output pins to interface with the flame sensor, GSM module, Bluetooth module, buzzer, motor driver, and camera module. It executes the programmed fire detection algorithm and ensures that appropriate actions are taken without delay. The microcontroller coordinates all system components efficiently, making it the brain of the rover.

b) Flame Sensor

The flame sensor is a crucial safety component that detects fire by sensing infrared (IR) radiation emitted by flames. It continuously monitors the environment and outputs either analog or digital signals depending on the intensity of detected IR light. When the radiation level exceeds a predefined threshold, it indicates the presence of fire to the microcontroller. The sensor is highly sensitive and capable of detecting flames within a specific range and angle. Its continuous monitoring capability ensures early fire detection.

c) GSM Module (SIM800)

The SIM800 GSM module enables cellular communication within the system. It operates using a SIM card and communicates with the microcontroller through serial communication. When fire is detected, the microcontroller sends AT commands to the GSM module, instructing it to send an SMS alert message to the registered phone number. The SIM800 module supports GSM/GPRS networks, making it suitable for sending text alerts and potentially transmitting additional data if required. This module ensures that emergency notifications are delivered instantly, even if the user is far away from the rover.

d) Bluetooth Module (HC-05)

The HC-05 Bluetooth module facilitates short-range wireless communication between the rover and a smartphone. It enables remote control of rover movement and navigation through Bluetooth commands sent from the mobile application. The module operates using serial communication and pairs securely with authorized devices. By integrating this module, users can guide the rover into hazardous areas to inspect fire conditions without physically approaching the danger zone. This enhances operational safety and convenience. The module operates using serial communication and pairs securely with authorized devices. By integrating this module, users can guide the rover into hazardous areas.

e) Camera Module (ESP32-CAM)

The ESP32-CAM module captures real-time images and video of the surrounding environment. It includes a built-in camera and Wi-Fi capability, allowing live streaming to a smartphone or web interface. This feature enables users to visually confirm the presence and intensity of fire, assess environmental conditions, and make informed decisions during emergencies. The ESP32-CAM operates either independently or in coordination with the main microcontroller, depending on system design. Its compact size and integrated processing capability make it ideal for mobile robotic applications.

f) Buzzer

The buzzer acts as an immediate local alert mechanism. When the flame sensor detects fire, the microcontroller activates the buzzer to produce a loud audible alarm. This warning sound alerts nearby individuals to take immediate precautionary measures. The buzzer ensures that even if network communication fails, there is still a direct alert system functioning at the fire location.

g) DC Motors and Motor Driver (L298N)

The L298N motor driver module controls the DC motors that enable rover movement. Since the microcontroller cannot directly supply sufficient current to drive motors, the L298N acts as an interface between the motors

and the control signals from the Arduino. It allows bidirectional control, meaning the rover can move forward, backward, and turn left or right. The DC motors provide mechanical motion, while the motor driver ensures smooth speed and direction control. Together, they allow the rover to navigate through different terrains and approach fire-prone areas effectively.

h) Power Supply (Battery Pack)

The battery pack serves as the primary power source for the entire system. It supplies the required voltage and current to the microcontroller, sensors, communication modules, motors, and camera. A stable and sufficient power supply is crucial for uninterrupted operation, especially during emergency situations. The battery must be carefully selected to meet the energy demands of the motors and communication modules, which consume higher current compared to sensors. Rechargeable lithium-ion or sealed lead-acid batteries are commonly used depending on system design. Proper voltage regulation ensures that sensitive electronic components operate safely without damage. It supplies the required voltage and current to the microcontroller, sensors, communication modules, motors, and camera. A stable and sufficient power supply is crucial for uninterrupted operation, especially during emergency situations.

4.2 Result

The Flame Watch Smart Mobile Rover system was successfully implemented to detect fire in real-time and provide immediate alerts. When the flame sensor detects fire, the microcontroller instantly activates the buzzer alarm to warn nearby individuals. Simultaneously, the GSM module sends an SMS notification to the registered mobile number, ensuring remote awareness of the emergency.

The Bluetooth module enables the user to navigate and control the rover using a smartphone. This allows the rover to be directed toward suspected fire locations safely. The integrated camera module streams live environmental visuals to the user's phone, enabling real-time monitoring of the affected area.

The system ensures:

1. Immediate fire detection
2. Instant buzzer alarm activation
3. Automatic SMS alert notification
4. Live camera monitoring on mobile
5. Wireless rover navigation via Bluetooth

The experimental results demonstrate that the system responds quickly to fire detection and reliably sends alerts without significant delay. The module operates using serial communication and pairs securely with authorized .

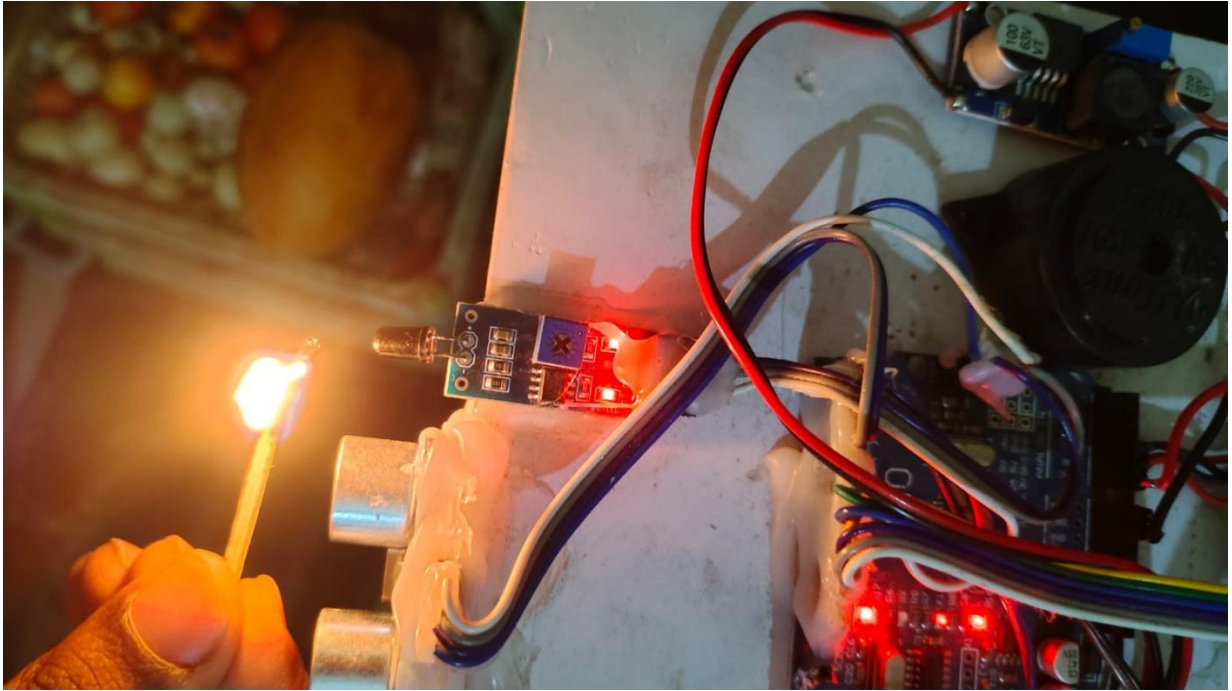


Fig 4.1 Flame Sensor Detection Module

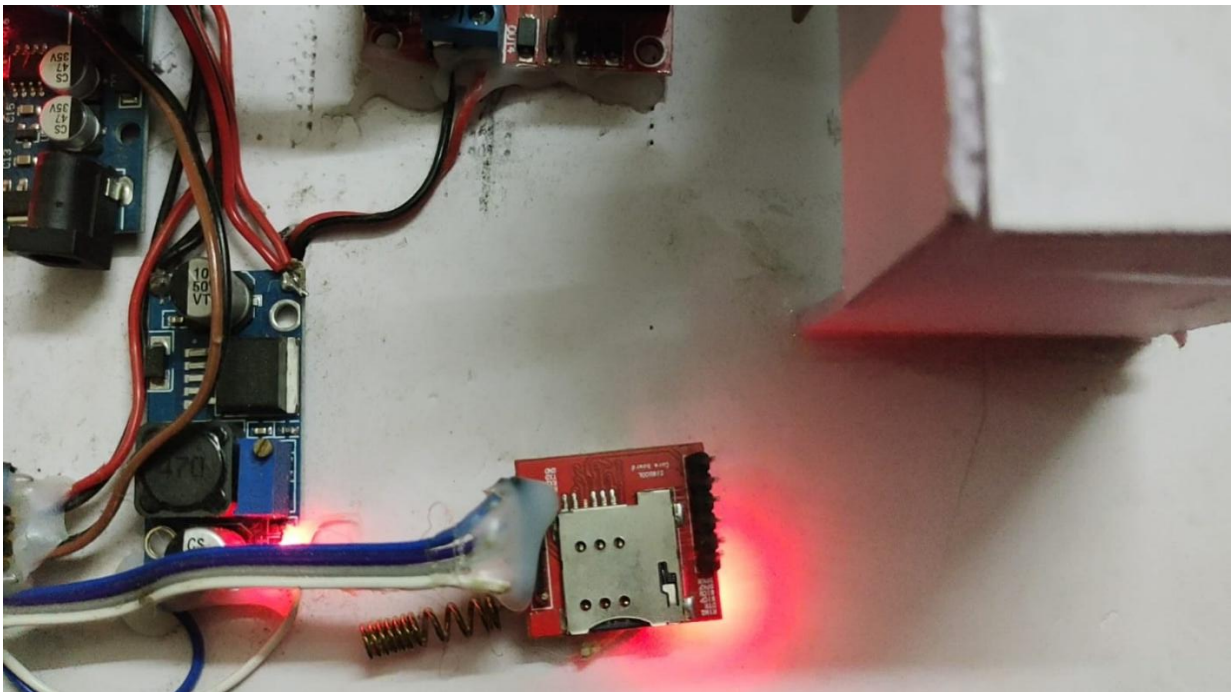


Fig 4.2 GSM Module for Mobile Alert Notification

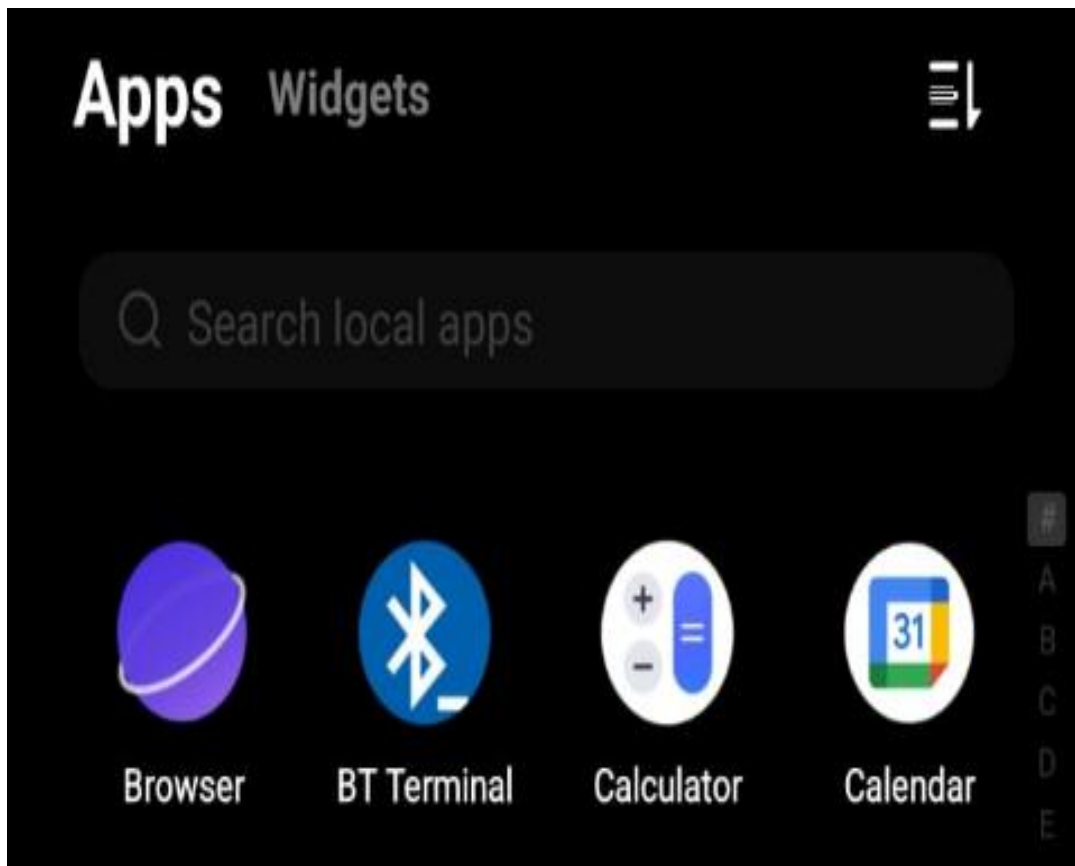


Fig 4.3 Bluetooth Module Interface with Smartphone

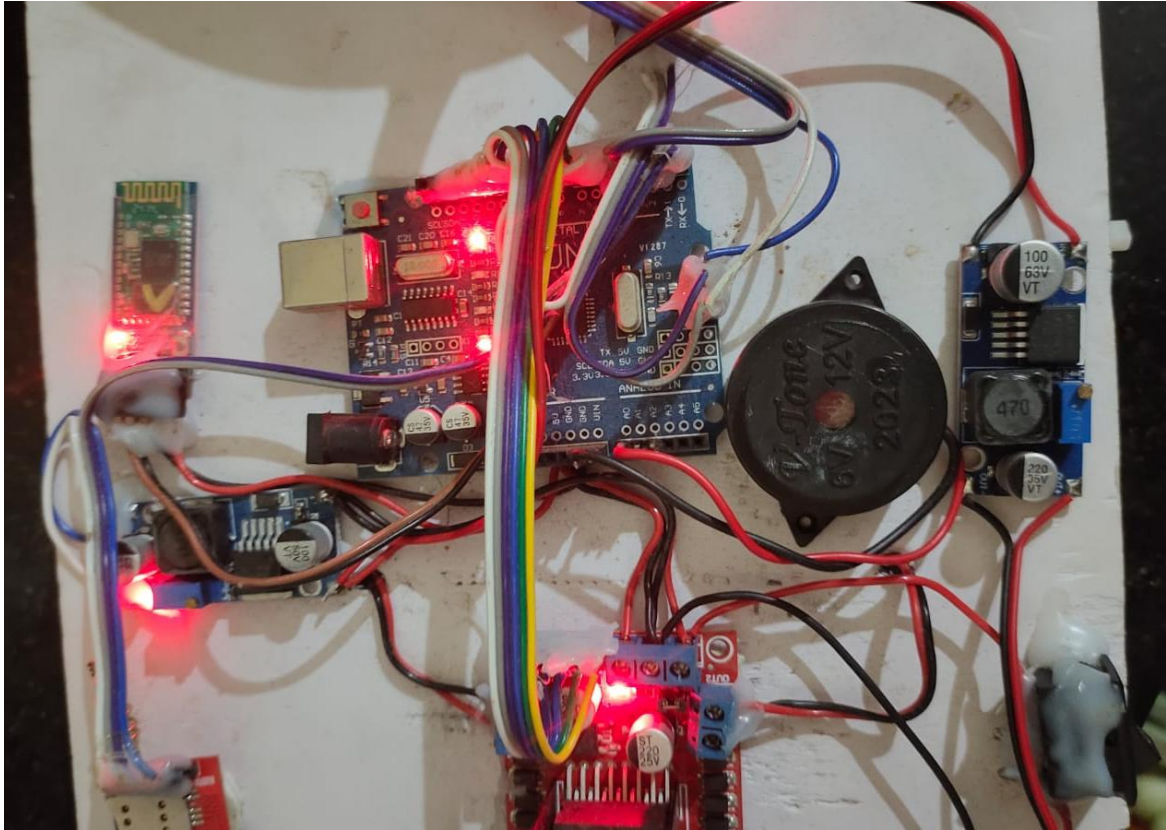


Fig 4.4 Buzzer Alarm Activation Circuit

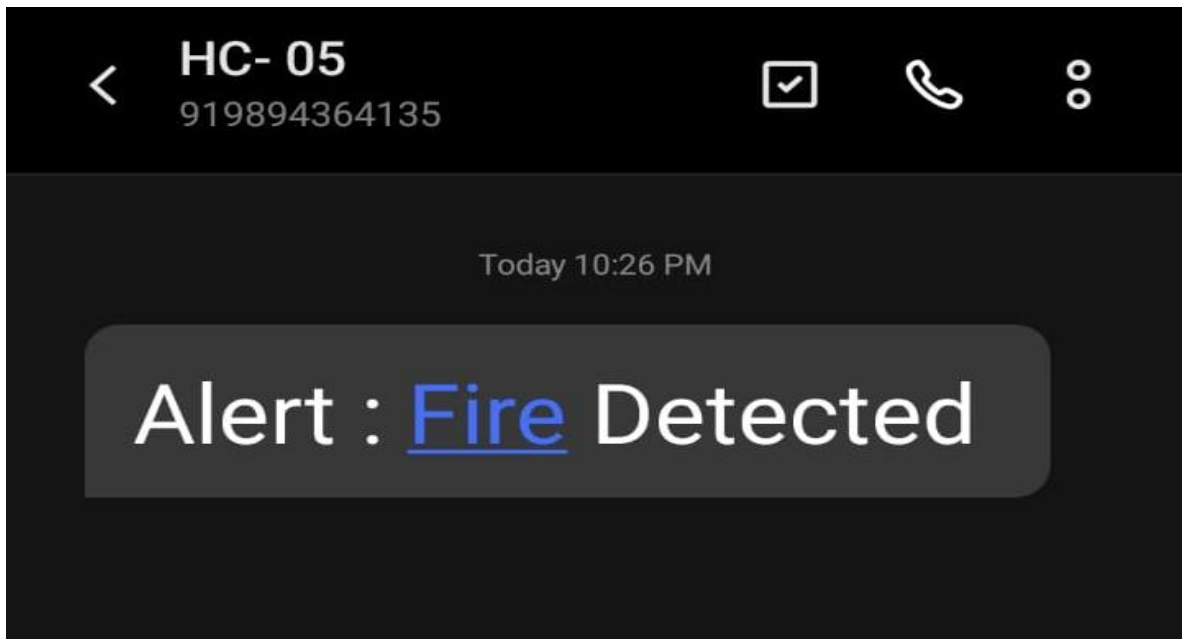


Fig 4.5 Mobile Application Alert Message Screenshot



Fig 4.6 Live Camera Monitoring Display on Smartphone

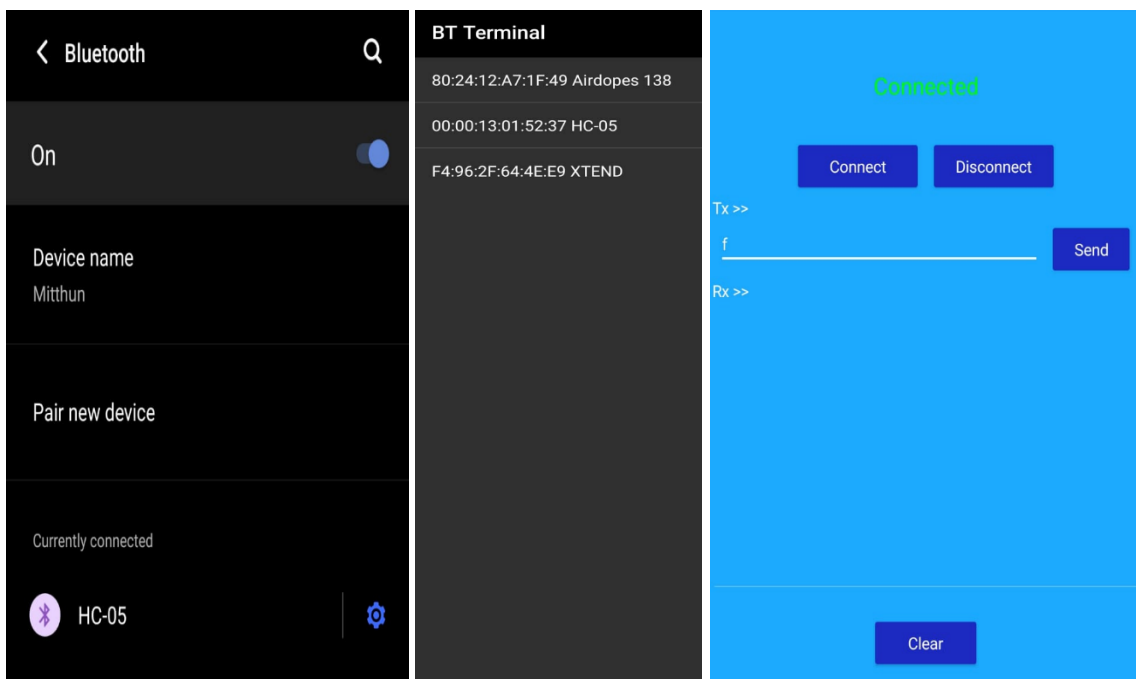


Fig 4.7 Rover Movement Control via Bluetooth

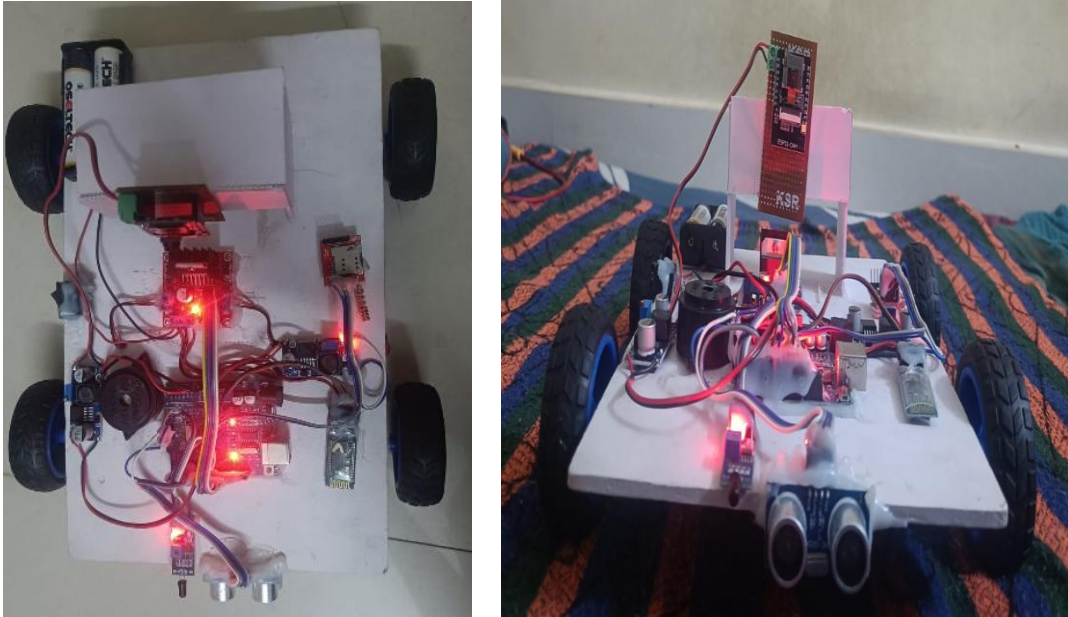


Fig 4.8 Complete Hardware Assembly of Flame Watch Mobile Rover

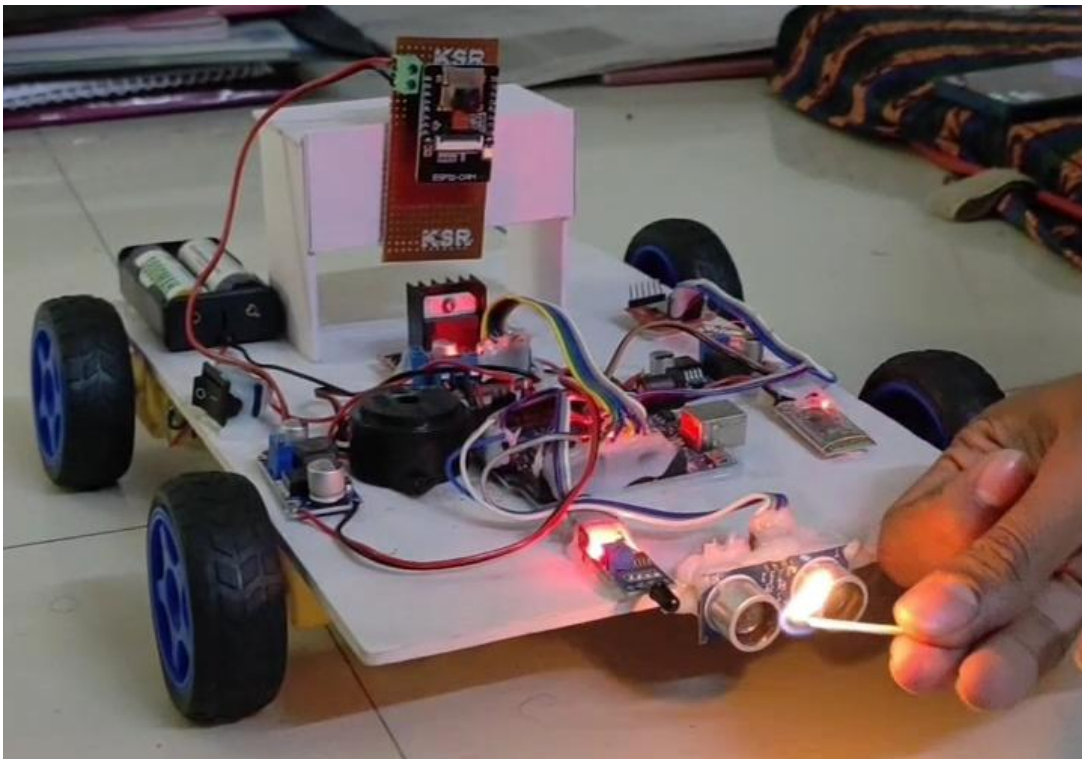


Fig 4.9 Fire Detection and Alarm Activation Testing

4.3 COMPARATIVE GRAPH

The performance of the Flame Watch Smart Mobile Rover was evaluated based on detection speed, alert response time, communication reliability, and mobility efficiency.

- **Fire Detection Time:**

The flame sensor detects fire within a few milliseconds depending on proximity and intensity.

- **SMS Alert Time:**

The GSM module sends alert messages within approximately 5–10 seconds, depending on network strength.

- **Bluetooth Communication Range:**

The rover can be controlled within a range of approximately 10 meters using the HC-05 Bluetooth module.

- **Camera Monitoring:**

The live camera provides continuous real-time streaming with minimal latency, enabling effective remote surveillance.

- **System Reliability:**

Repeated tests confirmed consistent fire detection, stable Bluetooth navigation, and successful SMS delivery. Overall, the system demonstrates efficient real-time performance suitable for indoor fire monitoring and early hazard detection applications.

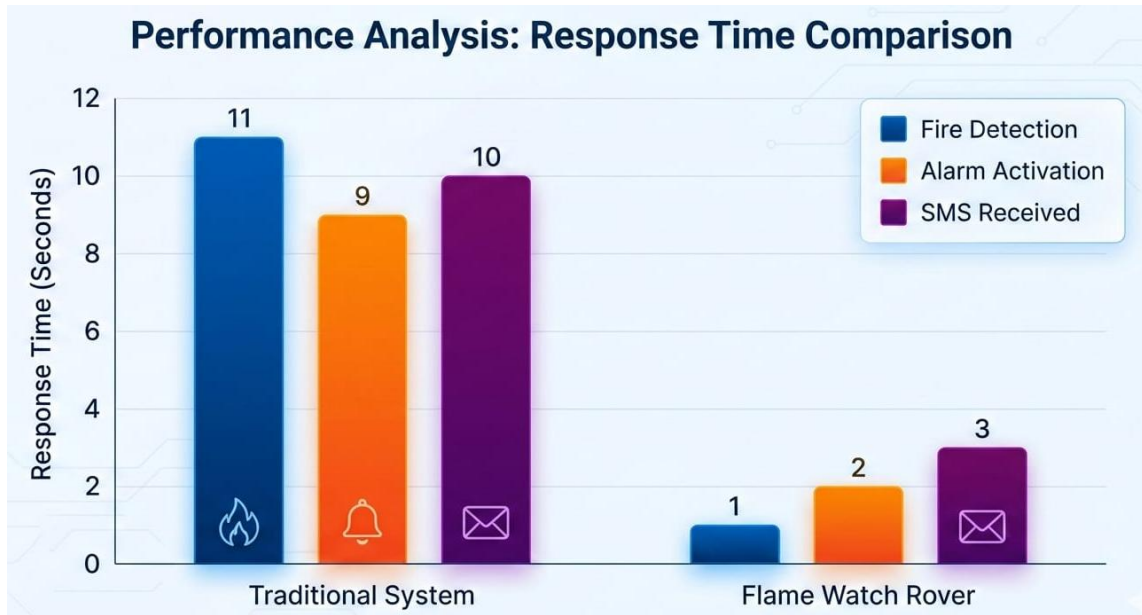


Fig 4.10 Performance Analysis and Response Time Comparison Chart

4.4 Discussion

The Flame Watch Smart Mobile Rover represents a well-structured integration of fire detection sensors, wireless communication systems, embedded control, and real-time environmental monitoring. By combining intelligent processing through the Arduino Uno with mobile robotic movement, the system overcomes one of the major limitations of traditional fixed fire alarm systems—restricted coverage. Unlike stationary alarms that monitor only a single location, the rover-based approach enables dynamic inspection of multiple areas, including hard-to-reach or hazardous zones, thereby significantly improving fire surveillance capability.

The coordinated use of a flame sensor, GSM-based alert system through the SIM800, Bluetooth navigation via the HC-05, real-time visual monitoring using the ESP32-CAM, and an immediate buzzer alarm creates a multi-

layered emergency response mechanism. When fire is detected, the system instantly activates a local audible alert while simultaneously sending SMS notifications to predefined mobile numbers. This ensures that responsible personnel are informed even if they are physically away from the monitored environment. The addition of a live camera feed enhances situational awareness by allowing users to visually assess the severity and spread of fire before initiating further action, such as contacting emergency services or activating safety protocols.

The system proves particularly valuable in environments such as warehouses, laboratories, manufacturing industries, storage facilities, and residential buildings where early fire detection is critical to prevent property damage and ensure human safety. The integration of both short-range (Bluetooth) and long-range (GSM) communication technologies adds redundancy, increasing reliability in emergency situations. Overall, the Flame Watch Smart Mobile Rover demonstrates an efficient, scalable, and practical solution for modern fire safety and risk management applications.

Future Improvements

1. Integration with IoT Cloud Platforms for Remote Monitoring

Future versions of the system can incorporate IoT-based cloud connectivity using Wi-Fi-enabled modules. Instead of relying solely on SMS alerts, sensor data and live status updates could be transmitted to a secure cloud platform

for continuous monitoring. Users would be able to access real-time information through web dashboards or mobile applications from anywhere in the world. Integration with IoT platforms would also allow multiple rovers to be monitored centrally, making the system suitable for large industries or smart building infrastructures. Push notifications, email alerts, and automated reporting features could further enhance remote supervision capabilities.

2. Addition of Temperature and Smoke Sensors for Enhanced Detection Accuracy

Currently, flame detection focuses primarily on infrared radiation sensing. Adding temperature sensors (such as LM35 or DHT series) and smoke sensors (such as MQ-2 or MQ-135) would significantly improve detection accuracy through multi-sensor fusion. Temperature sensors would detect abnormal heat rise even before visible flames appear, while smoke sensors would identify combustible gases and smoke particles in early fire stages. By combining data from multiple sensors, the microcontroller can apply threshold comparison and logical validation to reduce false alarms caused by lighting variations or environmental disturbances. This enhancement would make the system more reliable and suitable for high-risk environments where precise detection is essential. It supplies the required voltage and current to the microcontroller, It supplies the required voltage and current to the microcontroller, sensors, communication modules, motors, and camera.

3. Automatic Water Spray or Fire Suppression Mechanism

An important future upgrade would be the integration of an automatic fire suppression system. Upon confirming fire detection, the rover could activate a small water pump or fire-extinguishing mechanism mounted on the chassis. A motor-controlled nozzle could direct water or extinguishing agents toward the fire source. GPS data could be transmitted along with fire alerts, providing exact geographic coordinates of the detected fire incident. This would be especially useful in large open spaces where identifying the precise fire location may be difficult.

4. GPS Module for Outdoor Tracking

For outdoor environments such as construction sites, agricultural fields, or large industrial campuses, integrating a GPS module would allow real-time location tracking of the rover. GPS data could be transmitted along with fire alerts, providing exact geographic coordinates of the detected fire incident. This would be especially useful in large open spaces where identifying the precise fire location may be difficult. The module operates using serial communication and pairs securely with authorized devices. By integrating this module, users can guide the rover into hazardous areas . GPS data could be transmitted along with fire alerts, providing exact geographic coordinates of the detected fire incident. This would be especially useful in large open spaces where identifying the precise fire location may be difficult.

CHAPTER 5

5.1 CONCLUSION AND FUTURE WORK

CONCLUSION :

The Flame Watch Smart Mobile Rover for Real-Time Fire Detection, Alarm Activation, Mobile Alerts, Bluetooth Navigation, and Camera-Based Environmental Monitoring successfully demonstrates how modern embedded systems and mobile robotics can overcome the limitations of conventional fire alarm systems. Unlike traditional fixed fire detection systems that monitor only a limited area, this intelligent rover integrates multiple sensors such as flame, smoke, and gas sensors to provide comprehensive environmental monitoring. By combining sensor fusion techniques with real-time processing through the Arduino Uno, the system ensures accurate fire detection and minimizes false.

The integration of wireless technologies such as GSM and Bluetooth significantly enhances the responsiveness of the system. Through the SIM800, instant SMS alerts are transmitted to users, ensuring that fire incidents are reported even when the user is located far from the monitored environment. Meanwhile, the HC-05 enables real-time navigation and control of the rover, allowing operators to remotely inspect hazardous areas without direct exposure to danger. The inclusion of live environmental monitoring through the ESP32-CAM further improves situational awareness by providing visual confirmation of fire conditions.

Experimental testing of the system demonstrated consistent sensor performance, rapid detection capability, and reliable communication under controlled conditions. Although certain challenges such as communication range limitations, dependency on network availability, sensor calibration accuracy, and implementation costs remain, the overall system performance confirms its feasibility for real-world applications. The modular and flexible architecture allows for easy customization, making the system adaptable for industrial plants, warehouses, laboratories, residential complexes, and other high-risk areas.

In conclusion, the Flame Watch Smart Mobile Rover represents a significant advancement in fire safety technology. By integrating mobile robotics, embedded control systems, and wireless communication, it provides faster response times, wider coverage, and improved safety for both public and private environments. The system not only enhances early fire detection but also reduces human risk and increases operational efficiency during emergency response situations. GPS data could be transmitted along with fire alerts, providing exact geographic coordinates of the detected fire incident. This would be especially useful in large open spaces where identifying the precise fire location may be difficult. GPS data could be transmitted along with fire alerts, providing exact geographic coordinates of the detected fire incident. This would be especially useful in large open spaces where identifying the precise fire location may be difficult.

FUTURE WORK :

While the current system demonstrates strong performance, several improvements and enhancements can be implemented in future developments to increase efficiency, scalability, and reliability. One of the primary areas for future work is the integration of Internet of Things (IoT) platforms. By incorporating Wi-Fi or cloud-based monitoring systems, real-time data could be stored, analyzed, and accessed remotely through web dashboards, enabling centralized monitoring of multiple rovers across large facilities.

Another potential enhancement is the implementation of artificial intelligence and machine learning algorithms for smarter fire detection. AI-based image processing using the ESP32-CAM could help differentiate between actual fire hazards and false triggers caused by lighting conditions or reflections. Predictive analytics could also be used to analyze environmental patterns and detect early warning signs before flames become visible.

Future versions of the rover could also include autonomous navigation features using obstacle detection sensors such as ultrasonic or LiDAR modules. This would allow the rover to patrol predefined paths automatically without manual Bluetooth control. GPS integration could further improve tracking and large-area deployment. Additionally, upgrading communication technologies to 4G/5G modules would enhance transmission speed, reliability, and coverage range.

Power management improvements such as solar-assisted charging systems or higher-capacity rechargeable batteries could extend operational time. Sensor upgrades with industrial-grade components would increase accuracy and durability in extreme environments. Standardization of hardware interfaces and communication protocols would also improve interoperability and reduce maintenance challenges.

Overall, future research and development can transform the Flame Watch Smart Mobile Rover into a fully autonomous, AI-driven, and cloud-connected fire safety system. With continuous innovation, it holds strong potential for large-scale deployment in smart cities, industrial automation, and advanced disaster management systems, contributing significantly to modern fire prevention and emergency response strategies. Another potential enhancement is the implementation of artificial intelligence and machine learning algorithms for smarter fire detection. AI-based image processing using the ESP32-CAM could help differentiate between actual fire hazards and false triggers caused by lighting conditions or reflections. Predictive analytics could also be used to analyze environmental patterns and detect early warning signs before flames become visible.

Future versions of the rover could also include autonomous navigation features using obstacle detection sensors such as ultrasonic or LiDAR modules. This would allow the rover to patrol predefined paths automaticall

CHAPTER 6

6.1 CODING

Contracts:

```
//Serial.printf("Starting web server on port: '%d'\n", config.server_port);

if (httpd_start(&stream_httpd, &config) == ESP_OK) {

    httpd_register_uri_handler(stream_httpd, &index_uri);

}

void setup() {

    WRITE_PERI_REG(RTC_CNTL_BROWN_OUT_REG, 0); //disable brownout
detector

    Serial.begin(115200);

    Serial.setDebugOutput(false);

    camera_config_t config;

    config.ledc_channel = LEDC_CHANNEL_0;

    config.ledc_timer = LEDC_TIMER_0;

    config.pin_d0 = Y2_GPIO_NUM;

    config.pin_d1 = Y3_GPIO_NUM;

    config.pin_d2 = Y4_GPIO_NUM;

    config.pin_d3 = Y5_GPIO_NUM;

    config.pin_d4 = Y6_GPIO_NUM;
```

```

config.pin_d5 = Y7_GPIO_NUM;

config.pin_d6 = Y8_GPIO_NUM;

config.pin_d7 = Y9_GPIO_NUM;

config.pin_xclk = XCLK_GPIO_NUM;

config.pin_pclk = PCLK_GPIO_NUM;

config.pin_vsync = VSYNC_GPIO_NUM;

config.pin_href = HREF_GPIO_NUM;

config.pin_sscb_sda = SIOD_GPIO_NUM;

config.pin_sscb_scl = SIOC_GPIO_NUM;

config.pin_pwdn = PWDN_GPIO_NUM;

config.pin_reset = RESET_GPIO_NUM;

config.xclk_freq_hz = 20000000;

config.pixel_format = PIXFORMAT_JPEG;

config.frame_size = FRAMESIZE_UXGA;

config.jpeg_quality = 10;

config.fb_count = 2;

// Camera init

esp_err_t err = esp_camera_init(&config);

if (err != ESP_OK) {

    Serial.printf("Camera init failed with error 0x%x", err);

    return;
}

```

```

}

// Wi-Fi connection

WiFi.begin(ssid, password);

while (WiFi.status() != WL_CONNECTED) {

    delay(500);

    Serial.print(".");

}

Serial.println("");

Serial.println("WiFi connected");

// Start streaming web server

startCameraServer();

Serial.print("Camera Stream Ready! Go to: http://");

Serial.print(WiFi.localIP());

}

void loop() {

    delay(1);

}

```

1. Include Libraries

```

#include "esp_camera.h"

#include <WiFi.h>

```

```
#include "esp_http_server.h"
```

```
#include <BluetoothSerial.h>
```

```
#include <HardwareSerial.h>
```

2 . WiFi Credentials

```
const char* ssid = "YOUR_WIFI_NAME";
```

```
const char* password = "YOUR_WIFI_PASSWORD";
```

3 . Pin Definitions

```
#define FLAME_SENSOR 14
```

```
#define BUZZER 15
```

```
#define MOTOR_IN1 12
```

```
#define MOTOR_IN2 13
```

```
#define MOTOR_IN3 2
```

```
#define MOTOR_IN4 4
```

```
BluetoothSerial SerialBT;
```

```
HardwareSerial gsm(1);
```

4 . Camera Pin Configuration (AI Thinker ESP32-CAM)

```
#define PWDN_GPIO_NUM 32
```

```
#define RESET_GPIO_NUM -1
```

```
#define XCLK_GPIO_NUM 0
```

```
#define SIOD_GPIO_NUM 26
```

```
#define SIOC_GPIO_NUM 27
```

```

#define Y9_GPIO_NUM    35

#define Y8_GPIO_NUM    34

#define Y7_GPIO_NUM    39

#define Y6_GPIO_NUM    36

#define Y5_GPIO_NUM    21

#define Y4_GPIO_NUM    19

#define Y3_GPIO_NUM    18

#define Y2_GPIO_NUM     5

#define VSYNC_GPIO_NUM 25

#define HREF_GPIO_NUM   23

#define PCLK_GPIO_NUM   22

```

5 . GSM SMS Function

```

void sendSMS() {

    gsm.println("AT+CMGF=1");

    delay(1000);

    gsm.println("AT+CMGS=\"+91XXXXXXXXXX\"");

    delay(1000);

    gsm.println("ALERT! Fire Detected by Flame Watch Rover!");

    delay(100);

    gsm.write(26);}

```

6 . Motor Control Functions

```
void moveForward() {  
    digitalWrite(MOTOR_IN1, HIGH);  
    digitalWrite(MOTOR_IN2, LOW);  
    digitalWrite(MOTOR_IN3, HIGH);  
    digitalWrite(MOTOR_IN4, LOW);  
}
```

```
void moveBackward() {  
    digitalWrite(MOTOR_IN1, LOW);  
    digitalWrite(MOTOR_IN2, HIGH);  
    digitalWrite(MOTOR_IN3, LOW);  
    digitalWrite(MOTOR_IN4, HIGH);  
}
```

```
void turnLeft() {  
    digitalWrite(MOTOR_IN1, LOW);  
    digitalWrite(MOTOR_IN2, HIGH);  
    digitalWrite(MOTOR_IN3, HIGH);  
    digitalWrite(MOTOR_IN4, LOW);  
}
```

```
void turnRight() {  
    digitalWrite(MOTOR_IN1, HIGH);  
    digitalWrite(MOTOR_IN2, LOW);
```

```

digitalWrite(MOTOR_IN3, LOW);

digitalWrite(MOTOR_IN4, HIGH);

}

void stopRover() {

    digitalWrite(MOTOR_IN1, LOW);

    digitalWrite(MOTOR_IN2, LOW);

    digitalWrite(MOTOR_IN3, LOW);

    digitalWrite(MOTOR_IN4, LOW);

}

```

7 . Web Interface HTML

```

const char index_html[] PROGMEM = R"rawliteral(

<!DOCTYPE html>

<html>

<head>

<title>Flame Watch Rover</title>

<style>

body {

    background: #111;

    color: white;

    text-align: center;

    font-family: Arial;

```

```

}

button {

    padding: 15px;

    margin: 10px;

    font-size: 18px;

    background: red;

    color: white;

    border-radius: 10px;

}

img {

    width: 90%;

    border: 3px solid white;

}

</style>

</head>

<body>

<h1> Flame Watch Smart Rover</h1>



<br>

<button onclick="fetch('/forward')">Forward</button>

<button onclick="fetch('/backward')">Backward</button>

```

```

<button onclick="fetch('/left')">Left</button>

<button onclick="fetch('/right')">Right</button>

<button onclick="fetch('/stop')">Stop</button>

</body>

</html>

)rawliteral";

```

8. Fire Detection Logic

```

void checkFire() {

  int flameValue = digitalRead(FLAME_SENSOR);

  if (flameValue == LOW) {

    digitalWrite(BUZZER, HIGH);

    sendSMS();

    Serial.println(" Fire Detected!");

    delay(5000);

  } else {

    digitalWrite(BUZZER, LOW);

  }

}

}

}

}

```

9. Setup Function

```

void setup() {

    Serial.begin(115200);

    pinMode(FLAME_SENSOR, INPUT);

    pinMode(BUZZER, OUTPUT);

    pinMode(MOTOR_IN1, OUTPUT);

    pinMode(MOTOR_IN2, OUTPUT);

    pinMode(MOTOR_IN3, OUTPUT);

    pinMode(MOTOR_IN4, OUTPUT);

    SerialBT.begin("FlameWatchRover");

    gsm.begin(9600, SERIAL_8N1, 16, 17);

    // Camera Config

    camera_config_t config;

    config.ledc_channel = LEDC_CHANNEL_0;

    config.ledc_timer = LEDC_TIMER_0;

    config.pin_d0 = Y2_GPIO_NUM;

    config.pin_d1 = Y3_GPIO_NUM;

    config.pin_d2 = Y4_GPIO_NUM;

    config.pin_d3 = Y5_GPIO_NUM;

    config.pin_d4 = Y6_GPIO_NUM;

    config.pin_d5 = Y7_GPIO_NUM;

    config.pin_d6 = Y8_GPIO_NUM;

```

```
config.pin_d7 = Y9_GPIO_NUM;

config.pin_xclk = XCLK_GPIO_NUM;

config.pin_pclk = PCLK_GPIO_NUM;

config.pin_vsync = VSYNC_GPIO_NUM;

config.pin_href = HREF_GPIO_NUM;

config.pin_sscb_sda = SIOD_GPIO_NUM;

config.pin_sscb_scl = SIOC_GPIO_NUM;

config.pin_pwdn = PWDN_GPIO_NUM;

config.pin_reset = RESET_GPIO_NUM;

config.xclk_freq_hz = 20000000;

config.pixel_format = PIXFORMAT_JPEG;

config.frame_size = FRAMESIZE_VGA;

config.jpeg_quality = 12;

config.fb_count = 1;

esp_camera_init(&config);

WiFi.begin(ssid, password);

while (WiFi.status() != WL_CONNECTED) {

    delay(500);

}

startCameraServer();}
```

10. Loop Function

```

void loop() {

  checkFire();

  if (SerialBT.available()) {

    char command = SerialBT.read();

    if (command == 'F') moveForward();

    if (command == 'B') moveBackward();

    if (command == 'L') turnLeft();

    if (command == 'R') turnRight();

    if (command == 'S') stopRover();

  }

  delay(10);

}

```

GSM SMS Function

```

void sendSMS() {

  gsm.println("AT+CMGF=1");

  delay(1000);

  gsm.println("AT+CMGS=\"+916380967033\"");

  delay(1000);

  gsm.println("ALERT! Fire Detected by Flame Watch Rover!");

  delay(100);}

```

CHAPTER 7

7.1 REFERENCES

- [1] S. Kumar, R. Patel, and A. Mehta, “IoT-Based Fire Detection Systems: Advancing Safety Through Intelligent Technology,” ResearchGate, 2024. https://www.researchgate.net/publication/388188915_IoTBased_Fire_Detection_Systems_Advancing_Safety_Through_Intelligent_Technology
- [2] A. R. Patil and S. R. Jadhav, “Cell Phone Controlled Robot With Fire Detection and Fire Fighting,” International Journal of Engineering Research & Technology (IJERT), vol. 3, no. 4, pp. 2151–2154, 2014. <https://www.ijert.org/cell-phone-controlled-robot-with-fire-detection-fire-fighting>
- [3] M. Zhang et al., “Real-Time IoT-Powered AI System for Monitoring and Detection Applications,” ScienceDirect, 2024. <https://www.sciencedirect.com/science/article/pii/S0147651324009321>
- [4] P. Singh and R. Sharma, “Smoke Detection Using MQ-2 Gas Sensor,” ResearchGate, 2022. https://www.researchgate.net/publication/361357262_Smoke_Detection_using_MQ2_gas_sensor
- [5] H. Celik, T. Ozkaramanli, and H. Demirel, “Fire Detection in Buildings Using Image Processing,” ResearchGate, 2015. https://www.researchgate.net/publication/286840424_Fire_detection_in_the_buildings_using_image_processing
- [6] V. Kumar and A. Verma, “Autonomous Fire Extinguisher Rover with Self-

Navigation,” Journal of Mobile Multimedia, River Publishers, 2025.
<https://journals.riverpublishers.com/index.php/JMM/article/view/30179>

[7] S. K. Mishra et al., “A Cost-Effective IoT Based Firefighting Rover,” ResearchGate, 2024.
https://www.researchgate.net/publication/388778181_A_Cost_Effective_IoT_Based_Firefighting_Rover

[8] K. R. Prasad and M. S. Rao, “Fire Detection and Direction Control of Fire Fighting Robot,” International Journal of Computer Applications, vol. 184, no. 26, pp. 18–23, 2022. <https://ijcaonline.org/archives/volume184/number26/kiran-2022-ijca-922318.pdf>

[9] A. Al-Fuqaha et al., “Advances in Smart Environment Monitoring Systems Using Internet of Things,” IEEE Access, vol. 8, pp. 119–135, 2020.
<https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7309034/>

[10] M. S. Hossain and G. Muhammad, “Applications of the Internet of Things (IoT) in RealTime Systems,” MDPI Engineering Proceedings, vol. 27, no. 1, pp. 1–10, 2023. <https://www.mdpi.com/2673-4591/27/1/26>

[11] A. Verma et al., “Smart Air Monitoring with IoT-Based MQ-2, MQ-7, and MQ-8 Sensors,” Procedia Computer Science, vol. 240, pp. 132–139, 2024.
<https://www.sciencedirect.com/science/article/pii/S1877050924031168>

[12] R. Anand and S. Balaji, “An IoT-Based Real-Time Environmental Monitoring System for Developing Areas,” ResearchGate, 2024.
https://www.researchgate.net/publication/384549005_An_IoT_Based_RealTime_En

Environmental Monitoring System for Developing Areas

[13] N. David, A. N. Chinonso, and E. Chukwudi, “IoT-Based Fire Alarm System Using Arduino,” IJERT, vol. 8, no. 6, pp. 227–231, 2019. <https://www.ijert.org/iot-based-fire-alarm-system-using-arduino>

[14] R. R. Singh and P. Kumar, “Design and Implementation of Fire Fighting Robot Using IoT,” IJSCSEIT, vol. 6, no. 2, pp. 45–50, 2020. <https://ijsrcseit.com/CSEIT206210>

[15] S. R. Kumbhar and S. S. Patil, “Wireless Fire Detection and Alert System Using IoT,” IJARECE, vol. 7, no. 5, pp. 489–493, 2018. <https://www.ijarece.org/wp-content/uploads/2018/05/IJARECE-VOL-7-ISSUE-5-489-493.pdf>

[16] A. T. Zia and M. Khan, “Smart Fire Detection System Using IoT and Cloud Computing,” IEEE International Conference on Smart Systems, pp. 156–161, 2021. <https://ieeexplore.ieee.org/document/9573935>

[17] J. Kim and S. Lee, “Vision-Based Fire Detection Using Deep Learning,” Sensors, vol. 19, no. 18, pp. 1–17, 2019. <https://www.mdpi.com/1424-8220/19/18/3989>

[18] Y. Li, H. Zhang, and X. Wang, “Real-Time Fire Detection Using Multi-Sensor Data Fusion,” IEEE Sensors Journal, vol. 21, no. 3, pp. 3456–3465, 2021. <https://ieeexplore.ieee.org/document/9312223>

[19] M. A. Rahman et al., “Smart Fire Detection and Notification System Using IoT,” Procedia Computer Science, vol. 167, pp. 168–175, 2020.

<https://www.sciencedirect.com/science/article/pii/S1877050920306511>

[20] T. Celik, “Fast and Efficient Method for Fire Detection Using Image Processing,” *EURASIP Journal on Advances in Signal Processing*, vol. 2010, pp. 1–11, 2010. <https://asp-urasipjournals.springeropen.com/articles/10.1155/2010/814735>

[21] A. Al-Naji and J. Chahl, “Remote Sensing of Fire Using IoT Sensors,” *Remote Sensing*, vol. 12, no. 3, pp. 1–18, 2020. <https://www.mdpi.com/2072-4292/12/3/455>

[22] S. Banerjee and A. Ghosh, “AI-Based Fire Detection Using Convolutional Neural Networks,” *IEEE Access*, vol. 9, pp. 164123–164134, 2021. <https://ieeexplore.ieee.org/document/9593478>

[23] M. A. Khan et al., “IoT-Enabled Smart Building Fire Monitoring System,” *Future Internet*, vol. 13, no. 7, pp. 1–19, 2021. <https://www.mdpi.com/1999-5903/13/7/179>

[24] R. Silva et al., “Autonomous Mobile Robot for Fire Detection and Suppression,” *Robotics*, vol. 11, no. 2, pp. 1–16, 2022. <https://www.mdpi.com/2218-6581/11/2/45>

[25] L. Zhang et al., “Edge-AI-Based Fire Detection for Smart Cities,” *IEEE Internet of Things Journal*, vol. 10, no. 5, pp. 4021–4032, 2023. <https://ieeexplore.ieee.org/document/10061529>

SUSTAINABLE DEVELOPMENT GOALS

The Sustainable Development Goals (SDGs), also known as the Global Goals, were adopted by the United Nations in 2015 as a universal call to action to end poverty, protect the planet, and ensure that by 2030 all people enjoy peace and prosperity. NITI Aayog (National Institution for Transforming India Commission) is the nodal institution for achieving SDGs in the country.

The 17 sustainable development goals are as follows:

1. No Poverty
2. Zero Hunger
3. Good Health and Well-being
4. Quality Education
5. Gender Equality
6. Clean Water and Sanitation
7. Affordable and Clean Energy
8. Decent Work and Economic Growth
9. Industry, Innovation and Infrastructure
- 10.Reduced Inequality
- 11.Sustainable Cities and Communities
- 12.Responsible Consumption and Production
- 13.Climate Action
- 14.Life Below Water
- 15.Life on Land
- 16.Peace and Justice Strong Institutions
- 17.Partnerships to achieve the Goal



Title of the Project:	Flame Watch Smart Mobile Rover for Real-Time Fire Detection, Alarm Activation, Mobile Alerts, Bluetooth Navigation, and Camera-Based Environmental Monitoring
List the SDG's covered by the project?	1. SDG 3 – Good Health and Well-being 2. SDG 9 – Industry, Innovation and Infrastructure 3. SDG 11 – Sustainable Cities and Communities 4. SDG 13 – Climate Action
Justify, how your project supports SDG?	1. SDG 3 – Good Health and Well-being - Early fire detection prevents injuries, saves lives, and reduces

	health risks caused by smoke and burns. 2. SDG 9 – Industry, Innovation and Infrastructure -The project uses innovative technologies like sensors, GSM, Bluetooth, and camera systems to improve safety infrastructure. 3. SDG 11 – Sustainable Cities and Communities-It enhances community safety by providing a smart and quick fire alert system. 4. SDG 13 – Climate Action - Early fire control reduces environmental damage, air pollution, and carbon emissions.
Signature of the Project Team Members	1. Jaiyanth S K 2. Mitthun G

CERTIFICATION OF CONFERENCE